BAV99

BAV99 — High-Speed Dual Switching Diode

Manufacturer: Nexperia   Package: SOT-23

General Description

The BAV99 is a dual high-speed switching diode with a common cathode, housed in a SOT-23 package. It is used for high-speed switching, signal clamping and reverse polarity protection.

Features

- Repetitive reverse voltage VRRM = 70 V
- Forward current IF = 215 mA
- Reverse recovery time trr = 4 ns
- Common cathode configuration (pins 1 and 3 = anodes, pin 2 = cathode)

Pin Configuration

Pin assignment for the SOT-23 package (top view).

Pin	Name	Function
1	A1	Anode of diode 1
2	K	Common cathode
3	A2	Anode of diode 2

Absolute Maximum Ratings

Stresses beyond these values may cause permanent damage to the device.

Symbol	Parameter	Value	Unit
VRRM	Repetitive reverse voltage	70	V
VR	Continuous reverse voltage	70	V
IF	Forward current	215	mA
IFSM	Non-repetitive peak forward current	4.0	A
Ptot	Total power dissipation	250	mW

Electrical Characteristics

Typical values at TA = 25 degC unless otherwise noted.

Symbol	Parameter	Test condition	Min	Typ	Max	Unit
VF	Forward voltage	IF = 1 mA	-	0.65	0.80	V
VF	Forward voltage	IF = 50 mA	-	0.85	1.10	V
IR	Reverse leakage current	VR = 70 V	-	-	2.5	uA
trr	Reverse recovery time	IF = 10 mA	-	4	6	ns

Thermal Characteristics

Symbol	Parameter	Value	Unit
RthJA	Thermal resistance junction-ambient	500	K/W

Package Dimensions

Outline drawing for the SOT-23 package. All dimensions in millimetres.

Dimension	Min	Nom	Max	Unit
A (body length)	2.80	2.90	3.04	mm
B (body width)	1.20	1.30	1.45	mm

Ordering Information

Order code	Package	Marking	Packing
BAV99	SOT-23	A7	Reel, 3000 pcs
BAV99W	SOT-323	A7	Reel, 3000 pcs

Document revision 1.0. Values are typical at TA = 25 degC. Verify against the latest datasheet from Nexperia before design release.